

MC8P MIDI Controller - Build Guide

.axs instruments

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Bill of Materials

MC8P B.O.M	
Component	Qty
Teensy 4.0	1
Kailh Mechanical Switch	4
B100k Potentiometer	8
10k Resistor	4
220Ω Resistor	2
10nF Ceramic Capacitor	8
100nF Ceramic Capacitor	1
10uF Ceramic Capacitor	1
10uF Electrolytic Capacitor	1
100uF Electrolytic Capacitor	1
128x64 SSD1306 OLED Monochrome Display	1
PJ307 3.5mm Headphone Socket	1
DC-005 Barrel Jack socket	1
M3 10mm (Female-Female) Stand-offs	4
M3 10mm (Female-Male) Stand-offs	4
M3 Screws	4
M3 Nuts	4
M2 5mm (Female-Male) Stand-offs	2
M2 Screws	2
M2 Nuts	2
KailH Keycaps	4

MC8P B.O.M

(T18) Potentiometer caps

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Build Guide

Smaller components

- Solder resistors to the **Main Board**: R1–R4 (10k pull-down), R5 & R6 (MIDI Out).
- Solder ceramic capacitors: C1–C12 - **NOTE**: if electrolytic capacitors inhibit the front panel from fitting on properly, have them at a 90 degree angle.
- Install PJ307 headphone socket (J1).
- Install DC-005 barrel jack socket (J2)

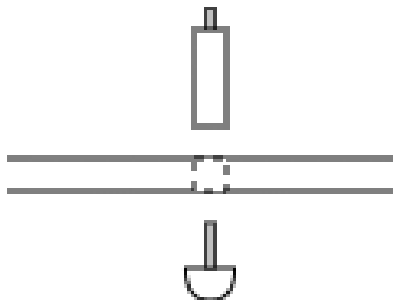
Teensy 4.0

- Solder header pins to Teensy.
- Mount and solder Teensy to the board.

OLED Display (SSD1306)

NOTE: The mounting screws for the OLED aren't necessary if the pin headers are long enough. You can solder the display in place with the ends of the pin headers being flush with the holes

- Flip board, insert 2× M2 screws through bottom holes.
- Attach female-to-male M2 stand-offs from front.
- Solder pin headers to the back of OLED.
- Align OLED with stand-offs and header holes.
- Secure with M2 nuts.
- Solder pin headers to **Main Board**





Switches & Potentiometers

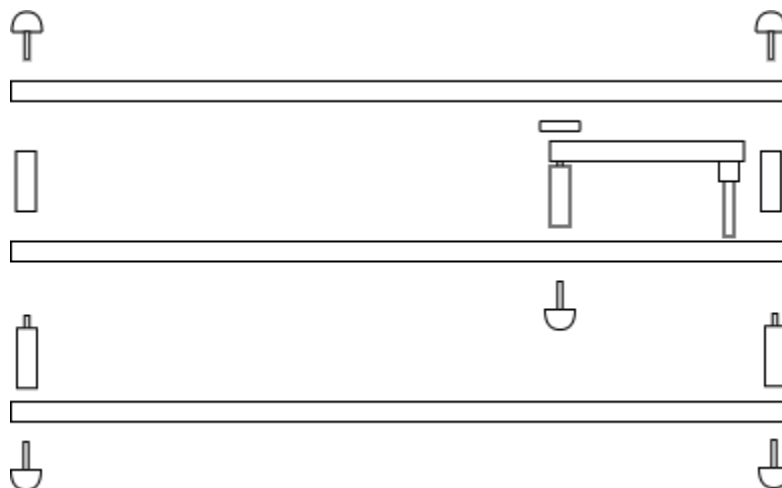
- Solder Kailh switches (SW1–SW4) to the **Main Board**.
- Solder potentiometers (RV1–RV8) to the Main Board.

Back Panel Assembly

- Insert 4× M2 screws (with nuts screwed on) through back panel holes.
- Attach 4× 10mm M3 female-male stand-offs to the blank side.
- Mount main board and secure with M3 female-female stand-offs.

Front Panel & Final Assembly

- Attach the front panel with remaining M3 screws + nuts.
- Secure potentiometers with washers and nuts.
- Add potentiometer knobs.



That's it!